Voice Deepfake Detection under Distribution Shift

Literature Review • Mayan Adler & Shlomi Hazan

1. The Problem and Established Countermeasures

Recent text-to-speech (TTS), voice conversion (VC), and neural codec-based generation methods produce increasingly convincing synthetic speech. Audio anti-spoofing aims to classify a recording as bona fide or spoofed, distinct from speaker verification, which tests a claimed identity. The central difficulty is not simply achieving low error on familiar recordings, but retaining discrimination when synthesis systems, speakers, corpora, or channel conditions change [1], [2]. A train–test generalization gap may exist even under a fixed distribution; out-of-distribution (OOD) evaluation additionally introduces a distribution shift. Poorer performance under such a shift does not, by itself, identify its cause. Formally, minimizing empirical loss on training data does not guarantee low expected risk under a different evaluation distribution:

$$R_{test}(h) = E_{(x,y)\sim P}\left[L(h(x),\, y)\right] \quad , \quad R_{test} \neq R_{train}$$

Here h is the detector, ℓ the loss, and P the joint distribution of audio and authenticity labels. This is a risk formulation, not the definition of EER.

ASVspoof 5 provides a benchmark with crowdsourced speech, adversarial attacks, and varied synthesis and channel conditions [1]. Its Track 1 evaluates standalone spoof detection using minDCF as the primary challenge metric, with equal error rate (EER) also reported; Track 2 evaluates spoofing-robust speaker verification using min a-DCF. In Track 1, an AASIST-based closed-condition baseline obtained 29.12% EER, versus 2.59% for an open-condition submitted system. These are different resource settings, not a controlled architectural ablation. Thus a headline score must be interpreted within its protocol.

CodecFake highlights a more specific shift toward neural codec-based synthesis and resynthesis [2]. A selected training configuration achieved 0.40% EER for VALL-E test examples, but this does not establish universal codec-attack detection. Importantly, neural compression of bona fide speech does not necessarily change its authenticity label, whereas codec-resynthesized outputs are designated spoof examples in CodecFake. A detector that learns a codec fingerprint alone could therefore reject legitimately processed speech.

Classical acoustic front ends such as MFCC, LFCC, and constant-Q cepstral coefficients (CQCC) search for spectral traces with predefined transforms. CQCC provides variable time–frequency resolution, yet its effectiveness varies with attack and corpus characteristics [3]. AASIST instead learns spectrotemporal dependencies with heterogeneous graph attention and reported 0.83% EER in its best run on ASVspoof 2019 LA (1.13% averaged across three seeds) [4]. Strong in-domain results remain insufficient evidence of transfer to newer attacks or datasets.

2. Learned Representations and Alternative Learning Paradigms

Self-supervised learning (SSL) offers a way to exploit large amounts of pre-training speech without requiring equally large spoof-labeled training sets. wav2vec 2.0 learns contextual representations via masked prediction and contrastive learning [5], while WavLM adds denoising-oriented pre-training and broader speech-task modeling [6]. Their original automatic speech recognition and speaker-related evaluations do not directly measure deepfake detection. Moreover, representation layers and temporal pooling can determine whether subtle synthesis traces survive aggregation; denoising could suppress nuisance variability but might also obscure relevant artifacts. Consequently, a frozen SSL encoder with a simple downstream classifier is a defensible test of representation quality, not an assumed solution to OOD spoofing.

Other studies seek to improve the information available to a detector. Wu et al. use guided VQ-VAE reconstruction to emphasize forensic traces; with log-power-spectrum features, EER on ASVspoof 2021 LA decreased from 22.64% to 14.04%, although improvements varied by representation and computational requirements increased [7]. Izrailov et al. combine MFCC and perceptual linear prediction with statistical and convolutional components, reporting 4.79% EER under their specified protocol [8]. The result does not isolate each component or establish an overall computational advantage. These approaches motivate controlled component comparisons rather than an assumption that added architectural complexity improves generalization.

A distinct strategy is to characterize authentic speech rather than learn the appearance of every spoof type. Yin and Lei combine wav2vec 2.0 and LFCC with Deep SVDD-based one-class components, obtaining 6.89% EER on ASVspoof 2021 DF compared with 18.45% for a reported supervised wav2vec 2.0 baseline [9]. However, differences in representations, fusion, and selection procedures prevent attribution of the entire gain to one-class learning. Crucially, labeled spoof samples appear in aspects of validation and tuning, so fitting an anomaly detector to real speech alone does not make the complete development pipeline spoof-label-free.

One-class detectors also face a conceptual trade-off: genuine recordings from unfamiliar microphones, languages, or speakers can lie outside the learned bona fide region, while convincing synthetic speech can lie within it. Their evaluation therefore must assess false rejection of real speech alongside missed spoofs. A meaningful comparison with binary classification should hold the speech representation fixed in advance and distinguish truly real-only training and model selection from procedures tuned with labeled fake validation data. This is a methodological control proposed for our study, not a claim established by the cited one-class paper.

Taken together, classical representations, graph-based deep detectors, SSL encoders, reconstruction, feature fusion, and one-class objectives exploit different cues. Their reported EER values are not a shared leaderboard: datasets, splits, labels, and resources differ. The research challenge is to compare representations and learning paradigms while changing as few other components as possible and to interpret failures under well-defined evaluation shifts.

3. Evaluation Validity, Dataset Diversity, and the Research Gap

Generalization is not a single experimental condition. Holding out synthesis systems (unseen-attack generalization), entire corpora (cross-dataset generalization), speaker identities (speaker generalization), and acoustic channels (channel robustness) tests different changes; several can occur simultaneously. A generator-held-out test may still share original recordings with training through other resyntheses, whereas reusing a known generator is not leakage if the claim concerns within-distribution performance. Strict claims require grouping related recordings and documenting what was actually held out [1], [2], [12].



Fig. 1. Distinct evaluation shifts (schematic, not measured results).



Fig. 2. AT-ADD Track 1 official baselines: Development versus Evaluation Macro-F1 [12].

Data diversity matters, but its causal interpretation is difficult. MLAAD v11 provides approximately 1,153.6 hours of synthetic speech from 205 TTS models in 54 languages; however, the article's reported cross-dataset comparisons used v1, so they do not demonstrate the incremental benefit of v11 or isolate sample count from generator variety [10]. PhonemeDF aligns parallel real and synthetic speech at phoneme level and shows that real–fake divergence is phoneme- and representation-dependent [11]. Correlations between divergence and classifier accuracy are often positive for spectral features but inconsistent for SSL embeddings. This suggests, without proving, that utterance-level averaging might miss local artifacts.

AT-ADD provides complementary benchmark evidence. Its Track 1 official FT-XLSR-AASIST baseline dropped from 99.70% Macro-F1 on development data to 76.73% on evaluation, while the leading challenge submission reached 90.71% on evaluation [12]. The evaluation combines previously unseen generators with other shifts; these results cannot isolate a single cause. Because Macro-F1 is calculated from class decisions rather than the EER of continuous detector scores, the two metrics are not numerically interchangeable. Per-generator spoof recall, bona fide recall by source, and uncertainty estimates are needed alongside aggregate results.

These findings define the gap addressed in our project: published improvements often modify the representation, classifier, training data, and evaluation conditions together, making it difficult to determine which component accounts for observed transfer. We will compare MFCC/CQCC and frozen wav2vec 2.0 under common splits, then evaluate binary versus genuinely real-only one-class learning with a representation fixed before spoof-labeled tuning. If provenance and access permit, a separate experiment will distinguish proportional increases in training-set size from allocating a fixed sample budget across more generators. The course subset contains ASVspoof 5 and CodecFake examples from both classes but only bona fide M-AILABS recordings; this creates potential dataset–label confounding and precludes an M-AILABS-only EER. Evaluation protocols, including whether unseen generators can truly be held out, will be fixed after metadata auditing and before model-result inspection. The aim is valid, reproducible evidence about when performance transfers—not an unsupported causal diagnosis of model behavior.

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